

An immune-aware benchmark of perturbation-prediction generalization

1 Tasks

5 curated, 3 classes
IMMUNE STIMULATION

T1 Kang IFN- β
PBMC · RNA

T2 Soskic 2022
CD4 T activation

CRISPR GENE

T3 Primary-T
CRISPR pool · RNA

T4 Frangieh
Perturb-CITE-seq

COMPOUND

T5 OP3
small molecules

2 Splits

held-out, leak-safe

Cell-context

held-out lineage
T1, T5

Perturbation

unseen target
T3, T4, T5

Modality

held-out readout
T4

Donor

held-out donor
T1, T2

 **Leak-safe**

Held-out cells appear
in the test fold only.
Response-gene panel,
PCA basis and floor are
fit on train cells only.

3 Methods

6 families + floor

 **Foundation**
reusable cell-state

 **Latent**
compositional effect

 **Prior-graph**
regulatory wiring

 **Optimal-transport**
distributional map

 **Hybrid**
repr. + transition op.

 **Chemistry-aware**
NO compound structure

ROLES

- **Conditioned predictors**
evaluated on held-out tasks
- **Deterministic predictors**
FP-ridge, linear-shift-KOemb
- **Comparator: CINEMA-OT**
never counted as a win

 **Universal floor**

cell-mean + linear-PCA shift

works only if it beats both members

4 Metrics

3 immune-aware

Response direction

Pearson- Δ
downstream Δ
 $\uparrow$ better

Program / readout

AUCCell- Δ
per-marker
 $\uparrow$ better

Distribution

E-distance
state-space
 $\downarrow$ better

5 Verdict

floor-adjudicated

Conditioned prediction

Beats BOTH floor members on the axes it supports?

cell-mean · linear-PCA

$\downarrow$ yes

 **Works on the task**

below floor

A split-half
reliability ceiling
bounds below-floor
results.